

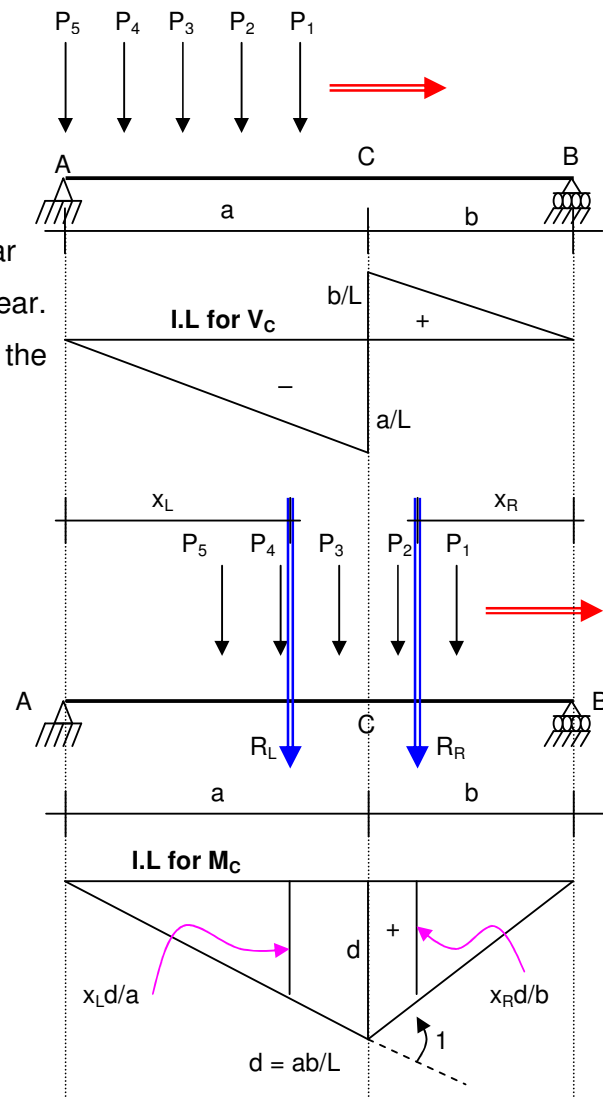
5.5 TRAIN OF POINT LOADS

5.5.1 General Observations

The maximum effects due to a train of point loads depend on the **magnitudes**, **position** and **spacing** of the loads, and the **span** of the beam. Consider a simply supported beam traversed by a train of point loads as shown alongside, and suppose we are interested in the shear force and bending moment at point C.

5.5.2 Max Shear force at C, V_C (max)

1. As long as P_1 has not arrived at C, V_C continues to cumulate and is negative.
2. When P_1 arrives just to the left of C, V_C attains its maximum negative value.
3. When P_1 passes C, it causes positive shear which reduces the cumulated negative shear.
4. The maximum positive shear occurs when the last load P_5 is just to the right of C.



5.5.3 Bending Moment at C, M_C (max)

1. Let P_1 and P_2 be to the right of C.
2. Let their resultant R_R act at distance x_R measured from B.
3. Let P_3 , P_4 , and P_5 be to the left of C.
4. Let their resultant R_L act at distance x_L measured from A.

Using the I.L. for M_C , we can write: -

$$M_C = R_L x_L d/a + R_R x_R d/b$$

$$= \{R_L x_L/a + R_R x_R/b\} d \quad \dots\dots\dots (1)$$

Suppose the train of loads moves a “small” distance to the right; then the change in moment at C may be evaluated as: -

$$\partial M_C / \partial x = (R_L/a - R_R/b) d \quad \dots\dots\dots (2)$$

Conclusions: Eqns 1 and 2 above show that:-

1. So long as $R_L/a > R_R/b$, then ∂M_C will be +ve, and M_C will continue to increase.
2. When P_3 has just passed C, **it may** make $R_L/a < R_R/b$. This will result in a –ve change in moment, and M_C will start to decrease. **[We say P_3 has tipped the balance]**
3. The maximum moment at C will occur when one of the point loads is just at the point C.
4. The point load that tips the balance must be **immediately on the left or on the right of the resultant of the train.**
5. The **absolute maximum moment** on the beam will occur under one of the point loads immediately to the left or right of the resultant of the train system, at a position to be determined in **article 5.5.4 below.**

5.5.4 Absolute Maximum Bending Moment and its Position

Suppose the resultant of the train of point loads in the fore-going illustrative example acts somewhere between P_3 and P_4 . Then the absolute maximum moment on the beam will occur “**somewhere**” when either P_3 or P_4 arrives “**there**”.

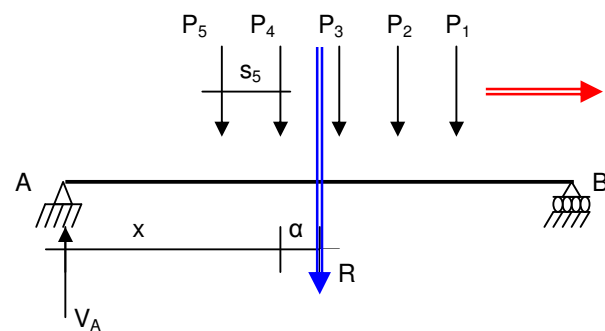
Scenario 1: Assume the absolute maximum moment is induced when P_4 arrives at the cross section X, distance x from A.

1. Let R be the resultant of all the point loads on the beam. $R = \sum P_i$
2. Let α be the distance of R from P_4 .
3. Let x be the distance of P_4 from A.
4. Let s_5 be the spacing of P_5 from P_4 .

Then equilibrium requires that: -

$$V_A = R(L - x - \alpha)/L$$

$$\text{and } M_4 = V_A x - P_5 s_5$$

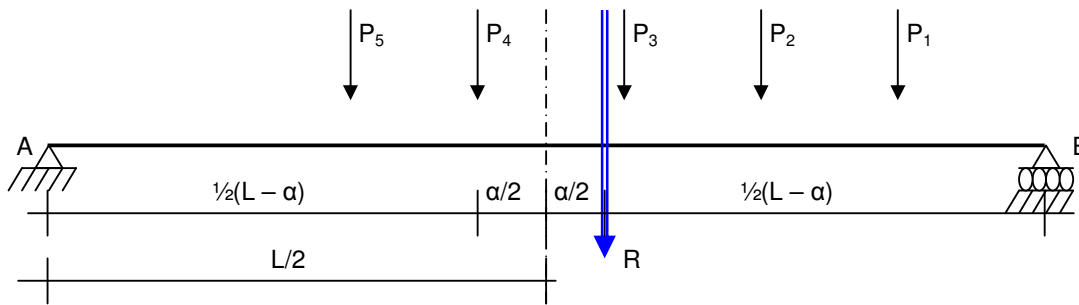


$$\text{So that } M_4 = R(Lx - x^2 - \alpha x)/L - P_5 s_5 \quad \dots\dots\dots (3)$$

The maximum moment under P_4 occurs when $\partial M_4 / \partial x = 0$

$$\partial M_4 / \partial x = R(L - 2x - \alpha)/L = 0 \implies x = \frac{1}{2}(L - \alpha) \quad \dots\dots\dots (4)$$

Scenario 1, represented by eqn 4, may be interpreted diagrammatically as shown below:



Notes

1. If R acts to the right of P_4 , $x = \frac{1}{2}(L - \alpha)$
2. If R acts to the left of P_4 , then $x = \frac{1}{2}(L + \alpha)$
3. If R coincides with P_4 , then $\alpha = 0$ and $x = L/2$

Conclusion

The absolute maximum moment under a point load will occur when the beam midspan bisects the distance between that particular point load and the line of action of the resultant of all the point loads on the beam.

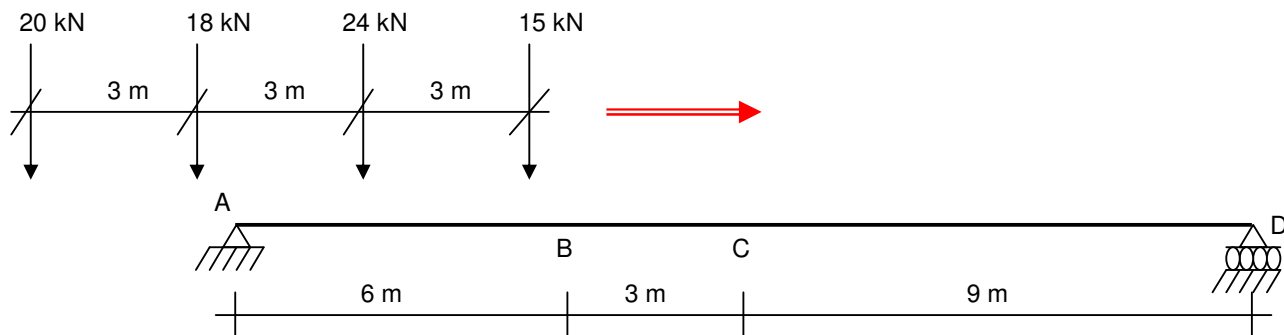
Notes

1. In sections 5.4 and 5.5 a simply supported beam was used to illustrate how to determine maximum load effects due to a udl segment and a train of point loads. The observations and conclusions arrived at are equally applicable to any beam span.
2. Once the maximum load effects have been determined using influence lines, the rest of the load effects, namely SFD, BMD, deflections and/or slopes may be evaluated, for that load scenario, using any of the methods covered earlier..

Example 7

A simply supported beam is traversed by a train of point loads as shown below.

- Determine the maximum shear forces induced at B and C.
- Determine the maximum bending moments induced at B and C.
- Determine the absolute maximum moment induced in the beam, stating where it is induced.

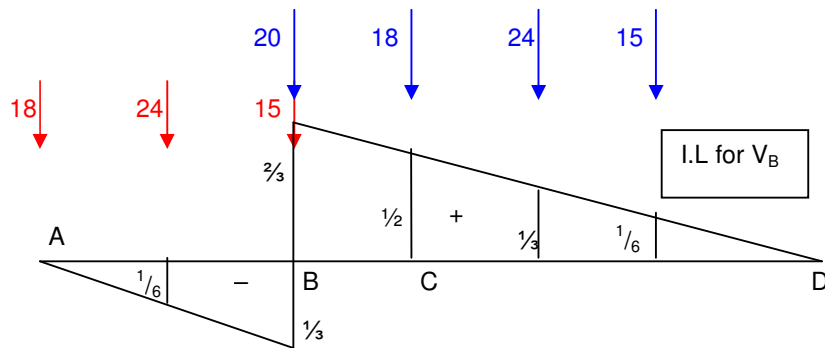


Solution

(a) Maximum shear forces

Using the I. L for V_B

Max. (-ve) shear at B occurs when the lead load is just to the left of B.



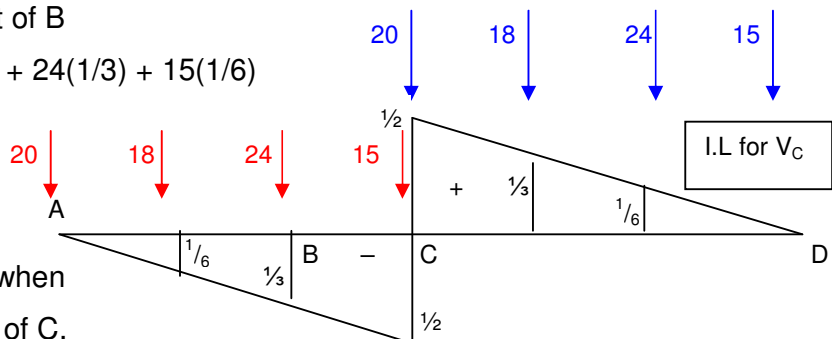
$$V_{B \text{ max (-ve)}} = 15(-1/3) + 24(-1/6) = -9.0 \text{ kN}$$

Max. (+ve) shear at B occurs when the tail load is just to the right of B

$$V_{B \text{ max (+ve)}} = 20(2/3) + 18(1/2) + 24(1/3) + 15(1/6) = 32.83 \text{ kN}$$

Using the I. L for V_C

Max. (-ve) shear at C occurs when the lead load is just to the left of C.



$$V_{C \text{ max (-ve)}} = 15(-1/2) + 24(-1/3) + 18(-1/6) = -18.5 \text{ kN}$$

Max. (+ve) shear at C occurs when the tail load is just to the right of C.

$$V_{C \text{ max (+ve)}} = 20(1/2) + 18(1/3) + 24(1/6) = 20.0 \text{ kN}$$

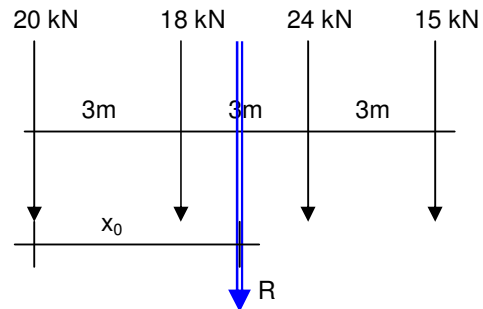
(b) Maximum bending moments

Let the resultant of the load system be at distance x_0 from the 20 kN tail load.

$$R = 20 + 18 + 24 + 15 = 77 \text{ kN}$$

$$x_0 \times 77 = 18 \times 3 + 24 \times 6 + 15 \times 9$$

$$\Rightarrow x_0 = 4.325 \text{ m}$$



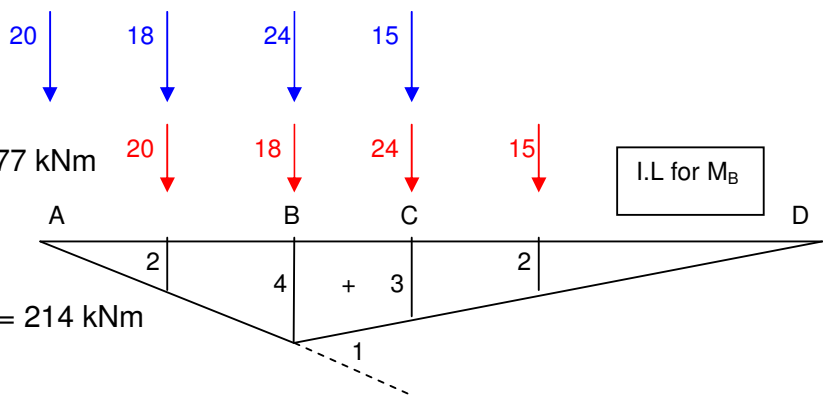
Max. moment at B occurs when either the 24 kN load or the 18 kN load arrives at the point B.

Scenario 1: 24 kN load at B

$$M_B = 18(2) + 24(4) + 15(3) = 177 \text{ kNm}$$

Scenario 2: 18 kN load at B

$$M_C = 20(2) + 18(4) + 24(3) + 15(2) = 214 \text{ kNm}$$



$M_B (\text{max}) = 214 \text{ kNm}$, when 18 kN load arrives at B.

Max. moment at C occurs when either the 24 kN load or the 18 kN load arrives at the point C.

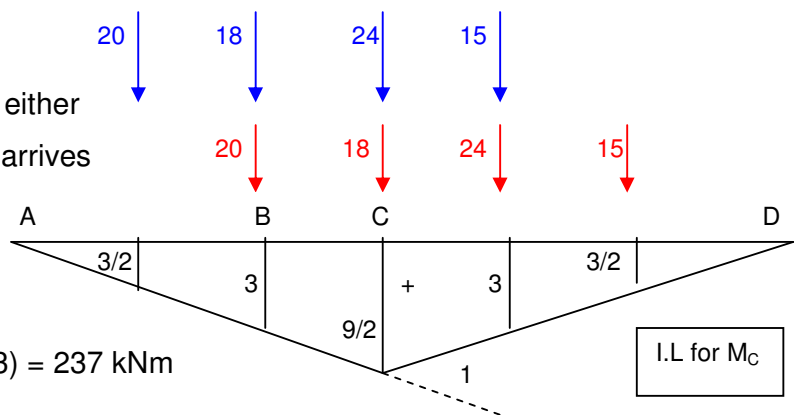
Scenario 1: 24 kN load at C

$$M_C = 20(3/2) + 18(3) + 24(9/2) + 15(3) = 237 \text{ kNm}$$

Scenario 2: 18 kN load at C

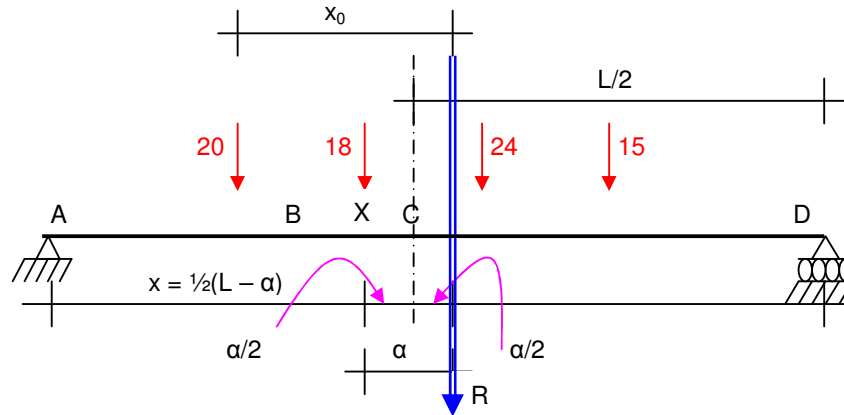
$$M_C = 20(3) + 18(9/2) + 24(3) + 15(3/2) = 235.5 \text{ kNm}$$

$M_C (\text{max}) = 237 \text{ kNm}$, when 24 kN arrives at C



(c) **Absolute maximum moment** will occur under either the 18 kN load or the 24 kN load.

Scenario 1: Assume the absolute maximum moment occurs under the 18 kN load:



Distance between 18 kN and R:

$$\alpha = x_0 - 3 = 1.325 \text{ m}$$

Position of 18 kN load from A:

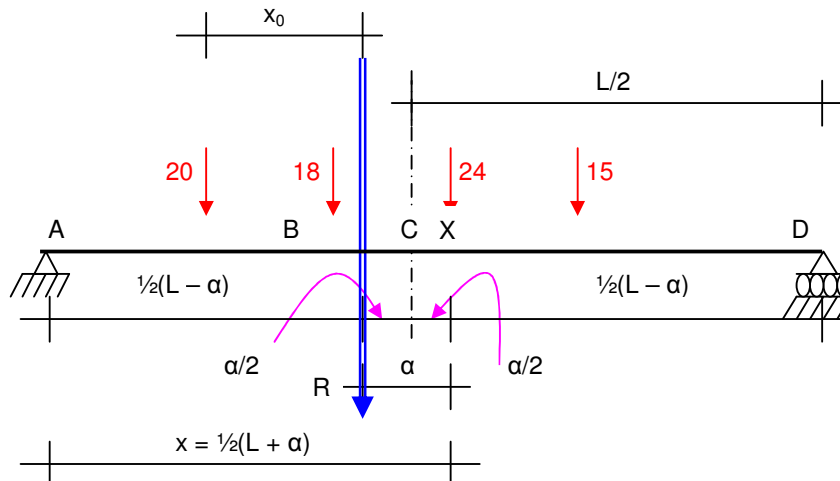
$$x = (1/2) (L - \alpha) = 8.338 \text{ m}$$

By equilibrium:

$$V_A = R (L - x - \alpha)/L = 35.664 \text{ kN}$$

$$M_{\max} = V_A x - 20(3) = 237.37 \text{ kNm}$$

Scenario 2: Assume the absolute maximum moment occurs under the 24 kN load:



Distance between 24 kN and R:

$$\alpha = 6 - x_0 = 1.675 \text{ m}$$

Position of 24 kN load from A:

$$x = (1/2)(L + \alpha) = 9.838 \text{ m}$$

By equilibrium:

$$V_D = R(x - \alpha)/L = 34.920 \text{ kN}$$

$$M_{\max} = V_D (L - x) - 15(3) = 240.02 \text{ kNm}$$

$$M_{(\text{abs}) (\text{max})} = 240.02 \text{ kNm}$$

which occurs at 9.838 m from A, when the 24 kN load arrives there.